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Alpilla 600 MW Combined Cycle Power Plant

Meteorological and Ambient Air Data

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	Prepared	Checked	Approved
Function	Owner's Engineer (discipline lead)	Owner's Engineer (project engineer)	Owner's Project Director
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CASE STUDY - all organisations, data and signatures are fictional.

1. Basis

Ten years (2015–2024) of hourly records from the Owner's on-site meteorological mast (installed 2014, 10 m and 60 m levels), checked against the nearest state meteorological station (22 km) and ERA5 reanalysis. Extremes use the state station's 1960–2024 record.

2. Air temperature (dry bulb, 2 m)

Month	Mean (°C)	Mean daily max (°C)	Mean daily min (°C)	Abs. max (°C)	Abs. min (°C)	Mean RH (%)
Jan	6.1	9.2	3.3	20.8	-11.4	80
Feb	6.6	10.1	3.5	23.1	-9.6	78
Mar	8.6	12.6	5.0	27.9	-6.2	77
Apr	12.5	16.9	8.6	30.4	-1.1	75
May	17.2	21.6	13.1	33.5	4.8	73
Jun	22.0	26.4	17.6	37.2	9.4	69
Jul	24.6	29.0	20.3	39.6	13.1	66
Aug	24.5	28.9	20.4	38.8	13.5	67
Sep	20.6	25.0	16.6	35.2	7.9	70
Oct	16.0	20.1	12.4	31.0	1.2	75
Nov	11.5	15.2	8.2	26.3	-4.3	78
Dec	8.0	11.3	5.0	21.7	-8.8	80
Year	14.9			39.6	-11.4	74

3. Design points (derived by the Owner)

Design point	Dry bulb (°C)	Coincident RH (%)	Remark
Summer 0.4 % exceedance	32.8	48	cooling / HVAC design
Summer 1 % exceedance	31.5	52	
Annual average	14.9	74	the contract Site Reference Condition uses 15.0 °C / 70 %
Winter 99 %	-2.6	85	heating / freeze protection
Winter 99.6 %	-4.8	88	
Record maximum	39.6	30	
Record minimum	-11.4	90	

Barometric pressure: annual mean 1011.5 mbar at site grade (+15.0 m); range 985–1038 mbar. Relative humidity range: 18 % (foehn, rare) to 100 % (fog). Fog: 21 days/year, mostly November–February.

4. Wind

Prevailing directions: NE (poyraz, 38 % of hours) and SW (lodos, 22 %). Lodos storms bring onshore wind with sea spray. Mean wind speed at 10 m: 4.6 m/s. Basic wind speed (10 min, 10 m, 50-year): 30 m/s. Maximum recorded gust: 38 m/s (SW, 2019).

5. Precipitation, snow, lightning

Annual precipitation 680 mm (max 24 h: 142 mm, 2021). Design intensity 110 mm/h (10 min, 25-year). Snow: 6 days/year on average, maximum recorded depth 34 cm; ground snow load 0.75 kN/m². Thunderstorm days: 26/year; ground flash density 2.4 flashes/km²/year.

6. Airborne salt and dust (measured by the Owner, 2023–2025)

Parameter	Annual mean	95th percentile	Maximum (event)	Remark
Sea-salt aerosol (as NaCl), at 10 m	14 µg/m ³	62 µg/m ³	165 µg/m ³	lodos storms, 2–4 events/year of 1–3 days
Sodium (Na) in PM10	4.5 µg/m ³	20 µg/m ³	54 µg/m ³	

Parameter	Annual mean	95th percentile	Maximum (event)	Remark
Chloride deposition (ISO 9225 wet candle)	180 mg/m ² ·day		420 mg/m ² ·day	ISO 9223 category S3
PM10	34 µg/m ³	78 µg/m ³	260 µg/m ³	Saharan dust episodes in spring
PM2.5	18 µg/m ³	41 µg/m ³	120 µg/m ³	

Atmospheric corrosivity per ISO 9223 is C5 (chloride-driven). The Owner considers salt ingestion into the gas turbine a major risk and expects the inlet air filtration to be designed for the event maximum above.

7. Performance test correction range

The Owner proposes that performance tests may be carried out at ambient temperatures between 5 °C and 30 °C; outside this range the test shall be postponed unless both Parties agree.